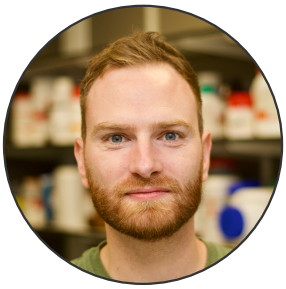


# Conor M. **Finlay**


Research Assistant Professor

*Trinity Translational Medicine Institute, Trinity College Dublin*

cofinlay@tcd.ie; conor.finlay@manchester.ac.uk | 0000-0001-8285-0903 | kQ6U7AkAAAAJ | conorisco | conorisco


*Immunologist*

## Research Roles

### Research Assistant Professor, Trinity College Dublin, 2025-present

My group investigates innate immune cells in inflammation, combining molecular immunology, in vivo science and computational approaches to understand immune cell function in health and disease.

| Role | Focus | Partners | Outputs |
|---|---|---|---|
| PI, Finlay Lab | Immune cell biology & bioinformatics | 3 PhD students | Macrophage biology, computational immunology |
| Acedemic Director, Trinity Omics | Single cell RNA-seq facility | TTMI-supported RA | Core technology service |
| Renal Research | Biomarker discovery | Prof. Mark Little, ADAPT | Data analytics, translational research |
| Pleural Disease Program | Macrophage function | Assoc. Prof. Laura Gleeson | Translational immunology |

### Honorary Research Fellow, University of Manchester (2021-present)

- Collaboration on *MRC* programme grant with Prof. Judith Allen. Pleural Macrophage biology.

## Past Employment History

**Snr Research Fellow, PI** — *Trinity College Dublin*
Mentor: Prof Mark Little — *2024-2025*
- Transitioned to research independence on SFI-Pathways award.

**Snr Research Fellow** — *Trinity College Dublin*
Supervisor: Aideen Long — *2023-2024*
- HEALED consortium - molecular biology and sequencing lead of tumour DNA and RNA-sequencing (with Aideen Long), since August 2023

**Research Fellow and subsequently Snr Research Fellow** — *Trinity College Dublin*
Supervisor: Prof Mark Little — *2021-2023*
- Single cell RNA-sequencing lead on COVID-19 strategic partnership. Laboratory management of Trinity Kidney Health Centre (THKC). Clinical data analysis, biomarker research, machine learning.

**Research Associate** — *University of Manchester*
Supervisor: Prof Judith Allen. — *2017-2021*
- Origin, heterogeneity, proliferation and effector function of macrophages during helminth infection and type 2 immune responses. Funding: Medical Research Council/Wellcome Trust. See recent preprint for outputs.

**Lecturer (teaching relief)** — *Trinity College Dublin*
School of Biochemistry and Immunology — *2016-2017*
- Performed the teaching duties for a senior professor on sabbatical for 1 year. Duties included undergraduate practical supervision, undergraduate and MSc teaching via lectures and tutorials, setting examinations and corrections with associated administration duties.

**Post-doctoral Researcher** — *Trinity College Dublin*
Supervisor: Prof Kingston HG Mills. — *2013-2016*
- For research projects see publications. I worked foremost on an IP-protected translational project identifying and testing novel helminth-derived proteins as therapeutics for inflammatory disease.

## Education

**Special Purpose Certificate in Teaching, Learning & Assessment for Academic Practice** — *Trinity College Dublin*
Perspectives on teaching and Learning in Higher Education — *2022*
- Constitutes one third of the SP Cert

**Informatics Training Scheme (Wellcome Trust TPA)** — *University of Manchester*
Programming and Computational Approaches to Biology Modules — *2020*
- Constituted 50% of taught compenent of MSc Bioinformatics and Systems Biology

**PhD in Immunology** *Trinity College Dublin*
Immune modulation by the helminth Parasite Fasciola hepatica *2013*
- Supervisor: Kingston Mills

**BA(Mod) in Natural Sciences** *Trinity College Dublin*
Major: Biochemistry with Immunology *2007*
- Grade: 1:1, graduated top of class

# Funding

## Awarded

**SFI-Pathways Programme** *Science Foundation Ireland*
594K, PI *2024-2028*
- Macrophage Activation and Transition in the Autoimmune Kidney (MacATAK): uncovering functional outcomes of macrophage differentiation in the inflamed kidney

**Trinity Doctorate Award 2024-2025, PI-led application** *Trinity Doctorate Award*
122K, PI *2024-2028*
- Mapping Macrophage Cell State Transition in Inflammation and Infection at Single Cell Resolution

**The Meath Foundation 2024** *The Meath Foundation*
70K, Co-I (35K to my group) *2024-2026*
- 'Characterisation and Inhibition of Monocyte-Driven Inflammation in AAV (CIMDIA)'

**Higher Education Research Equipment Grant** *Higher Education Authority*
30.2K, PI *2024-2025*
- Cell counting solution to support TCD Omics

**Dean's Research Initiatives Fund** *Faculty of Health Sciences, TCD*
20K, PI *2024*
- Mapping human macrophage anti-microbial function in pleural infection

**Enterprise Ireland-funded Industry collaboration** *Legend Biotech*
93K *2024*
- Title withheld, scRNA-sequencing of blood cancer

**Building Engagements in Health Research Scheme** *Internal TCD*
10K Eur, Co-PI *2023*
- Understanding mast cells and their progenitors in Telangiectasia Macularis Eruptiva Perstans (TMEP)'

**MR/V011235/1 Programme Grant** *Medical Research Council*
2.26 Million GBP, Research Co-Investigator *2021*
- "Macrophages in type 2 immunity: unravelling susceptibility and resistance to tissue nematode infection". I co-wrote grant and provided scientific design and preliminary data. Awarded an honorary position at UoM to facilitate ongoing collaboration after leaving Manchester

**Building Engagements in Health Research Scheme** *Internal TCD*
10K Eur, Co-PI *2021*
- 'Nanoparticle modulation of neutrophil and monocyte responses to ANCA'

## Under review

- 2025 HRCI HRB 300K, outcome July 2026
- 2025 HRB ILP 430K, outcome July 2026

# Presentations, Awards & contributions

## Oral Presentations

| | | |
|---|---|---|
| 2024 | UK Kidney Week (Invited talk) | *Edinburgh* |
| 2023 | AI-driven Digital Content Technology (ADAPT) Collaboration Day (Invited talk) | *TU Dublin* |
| 2023 | Irish society of Parasitology | *Tralee, Ireland* |
| 2023 | Building Engagements in Health Research | *TCD, Dublin* |
| 2022 | Mononuclear Phagocytes in Health and Disease (Invited talk) | *CRICK, London* |
| 2022 | Parasitic Helminths: New Perspectives in Biology and Infection | *Hydra, Greece* |
| 2022 | COVID-19 Immunology, Vaccines and Lessons for the Future (Invited talk) | *TCD, Dublin* |
| 2022 | Trinity Translational Medicine Institute - Conference 2022 (Invited talk) | *TTMI, Dublin* |
| 2021 | Cytokine and interferon society meeting | *Cardiff* |
| 2019 | Irish Society of Immunology (best presentation award) | *RCSI, Dublin* |
| 2019 | BSI Type 2 Immunology Meeting | *Manchester* |
| 2019 | KU Leuven, UZ Gasthuisberg Campus (Invited talk) | *KU Leuven* |
| 2015 | Molecular and cellular biology of helminth parasites IX | *Hydra, Greece* |
| 2015 | TBSI Post-Doc Research Day | *TCD, Dublin* |
| 2015 | 1st TBSI-Weimann joint Immunology conference | *Rehovot, Israel* |
| 2015 | European Congress of Immunology | *Vienna* |
| 2009 | European Congress of Immunology | *Berlin* |

## Poster Presentations

| | | |
|---|---|---|
| 2023 | British Society of Immunology | *Belfast* |
| 2019 | British Society of Immunology | *Liverpool* |
| 2018 | Type 2 Immunity in Homeostasis and Disease | *Bruges* |
| 2017 | British Society of Immunology | *Brighton* |
| 2016 | British Society of Immunology | *Liverpool* |
| 2015 | Frontiers in neurology | *TCD, Dublin* |
| 2014 | Cytokine and interferon society meeting | *Melbourne* |
| 2012 | Keystone: Biology of Cytokines | *Colorado* |
| 2012 | British Society of Immunology | *Colorado* |
| 2011 | Cytokine and interferon society meeting | *Florence* |

## Media and Commentary

- Commentary in *Immunity*: “Macrophages show up in style when Th2 lymphocytes organize their homecoming”
- Paper featured on The Immunology Podcast - Episode 51
- Immunity paper featured on August 2023 edition of *Journal of Allergy and Clinical Immunology* News Beyond Our Pages column.
- Irish Health Professional 2023
- Medical Express 2023

## Awards

| | |
|---|---|
| 2022 | Wiley Top Cited Article Award - Parasite Immunology |
| 2021 | EFIS/EJI Travel/Abstract Award Winners Presenting at Cytokines 2021 |
| 2019 | Best presentation from selected abstracts - Irish Society of Immunology |
| 2015 | Visiting student training scheme - Weizmann Institute of Science |
| 2015 | TBSI Post-Doc Research Day - Runner up talk prize |
| 2014 | Milstein Travel Award - International Cytokine society |
| 2007 | Valedictorian - Biochemistry with Immunology degree |
| 2007 | Best Undergraduate Research Poster Prize - School of Biochemistry and Immunology |

## Discipline Contributions

| | | |
|---|---|---|
| 2026 | Kidney Research UK Clinical fellowships | *Grant review* |
| 2026 | Kidney Research UK project grants | *Grant review* |
| 2025 | Nature Communications | *Article review* |
| 2025 | PhD studentship (Redacted UK Cancer Charity) | *Grant review* |
| 2024 | European Journal of Immunology | *Article review* |
| 2024 | European Congress of Immunology | *Poster session reviewer* |
| 2024 | European Congress of Immunology | *Session Chair* |
| 2023-2025 | JSAN | *Article review* |
| 2023 | Kidney Research UK | *Grant review* |
| 2023 | Annals of the Rheumatic Diseases | *Article review* |
| 2023 | British Fellowship awarding body (name withheld) | *Grant review* |
| 2022 | Irish Nephrology Society Conference | *Session Chair* |
| 2022 | Post-doctoral scientist at TCD | *Interview panel* |
| 2022 | PhD Candidate in TCD | *Continuation Viva* |
| 2022 | Kidney International | *Article review* |
| 2021-2022 | Frontiers in Immunology | *Article review* |
| 2021-2022 | Research Assistant at TCD | *Interview panel* |
| 2021 | French National Research Agency (ANR) 2021 generic call | *Grant review* |
| 2021 | BBSRC Discovery Fellows award | *Grant review* |
| 2021 | Venture capital funded Trinity Spinout – Parvalis Tx | *Patent (inventor)* |
| 2018 | Scientific Reports | *Article review* |
| 2017-2020 | Parasite Immunology | *Article review* |

## Active Collaborations

*Past Collaborations listed in Publications*

**Kathy Gately** — *TCD, St James' Hospital*
Malignang Pleural Effussion

**Luara Gleeson** — *TCD*
Human Pleural biology

**Padraic Fallon** — *Trinity College Dublin*
Title withheld; inflammation focused (mouse)

**Arthur White** — *Trinity College Dublin*
New statistical methods for semi-supervised clustering of flow and mass cytometry data

**Legend Biotech, Tony Mc Elligott & Nina Orfali** — *TCD*
Title withheald

**Judi Allen, Lili Zhang** — *University of Manchester*
B cell-macrophage interactions in type 2 immune responses

**Niall Conlon, Katie Ridge** — *St James' Hospital*
Role of mast cell progenitors in chronic urticaria

**Ross MacManus** — *Trinity College Dublin*
Linking immune cell perturbations to ankylosing spondylitis outcomes

**Mark Travis** — *University of Manchester*
Title withheld, T cell and infection focused (mouse)

**Sharee Basdeo** — *Trinity College Dublin*
Innate immune training of human monocytes and macrophages by SARS-Cov2 vaccine ChAdOx1

**Judi Allen** — *University of Manchester*
Serous cavity macrophage in nematode control

**Judi Allen, James Parkinson** — *University of Manchester*
The pleural cavity macrophage proteome

**Alvaro Diaz** — *Uni. de la República Uruguay*
Role of CD40 signalling in tissue resident macrophage proliferation

## Teaching & Supervision

### Formal teaching

**Introduction to bioinformatics, transcriptomics and single cell RNA-sequencing (Full week)** — *30 Students*
MSc in Molecular Medicine — *2026*

- Week long course of workshops, lectures and practicles. 18 hours of direcet teaching time.

**Intro to data science** *25 Students*
MSc Healthcare Infection *2026*

**RNA-sequencing (3 Lectures)** *25 Students*
MSc in Genomic Medicine *2025*

**Cell Atlases** *25 Students*
MSc in Molecular Medicine, Genomic Medicine *2025*

**Trinity Single Cell Analysis Workshop (Monthy Meeting)** *12 Members*
Open to Postgraduates and Post-Docs *2025*

**Heritability, Genotyping, Genetics of Heritable Disease** *Medium class size*
Bioengineering (4th year) *2024*
- Explores genetic inheritance, disease patterns, and population genetics. Students learn heritability concepts, genotyping methods, and genetic disease mechanisms, combining theoretical principles with practical medical applications.

**Malaria** *150 students*
Senior Freshmen Biology *2024*

**Introduction to omics technologies** *20 students*
Core Research Skills Module *2023*
- General introduction to omics technologies to new PhD students

**Transcriptomics (2 hr workshop)** *20 students*
MSc in Immunology *2022*
- Introduction to Transcriptomics and single cell technologies

**Datascience for Immunology (3 workshops and assessment)** *20 students*
MSc in Immunology *2022*
- 12 hours direct teaching time. Introduction to R programming, with a focus on analysis in R markdown. Analysing and visualising immunology data in R. Single cell RNA-sequencing analysis. Student assessment in Single cell RNA-sequencing analysis. Module design and module reform.

**R programming, data visualisation and transcriptomic analysis (3 workshops)** *20 students*
MSc in Immunology *2021*
- Part of immunogenetics module - This included 3 dry lab sessions (in person) and assessment: student presentations of bioinformatic analysis (microarray)

**T cells (4 lectures)** *30 Students*
BA(Mod) Biochemistry with Immunology (4th year) *2016-2017*
- This advanced course teaches most aspects of T cell biology from development, T cell education, tolerance, activation through the immunological synapse, APC function, co-stimulation, expansion, tissue/lymph node homing, T cell subset differentiation, regulation and the concept of immune responses types. Assessment: written exam

**Cancer immunology and immunotherapy (2 lectures)** *30 Students*
BA(Mod) Biochemistry with Immunology (4th year) *2016-2017*
- Brief history of chemotherapy, failed early immune oncology interventions, recent success, checkpoint inhibitors, combination studies, CAR-T cell, future perspectives Assessment: written exam

**Multiple sclerosis (1 lecture)** *30 Students*
BA(Mod) Biochemistry with Immunology (4th year) *2016-2017*
- Th1 and Th17 cells in autoimmunity, animal models of disease, S1p1 inhibitors, VLA4, future targets.

**T cell differentiation, effector function and regulation (2 lectures)** *20 students*
MSc in Immunology *2016-2017*
- Introductory course to T cell biology, differentiation and the cytokine control of immune responses. Assessment: written exam

**Introduction to parasite and type 2 Immune responses (3 lectures)** *20 students*
MSc in Immunology *2014-2017*
- This is an advanced course that covers aspect of type 2 immune responses. Th2 cells, helminth infections, eosinophils, M2 macrophages, wound repair, hygiene hypothesis and allergy. Workshop style with three PBL topics. Assessment: written exam

**Integrated Tutorials** *3 students*
BA(Mod) Biochemistry with Immunology (3rd year) *2014-2017*
- Ongoing small group tutorials for 3 students each year integrated with specific modules

## Research Project Supervision

| | | |
|---|---|---|
| 2026 | MSc Genomic Medicine | *TCD* |
| 2026 | MSc Molecular Medicine | *TCD* |
| 2026 | Medicine (Summer scholarship) | *TCD* |
| 2026 | Biomedical & Molecular Diagnostics (Placment) | *Technological University of Dublin* |
| 2025 | BA(mod) Human Health and Disease | *TCD* |
| 2025 | MSc Genomic Medicine | *TCD* |
| 2025 | BA(Mod) Molecular Medicine | *TCD* |
| 2025 | BA(Mod) Molecular Medicine | *TCD* |
| 2024-2028 | PhD Immunology (sole-supervisor) | *TCD* |
| 2024-2028 | PhD Bioinformatics (sole-supervisor) | *TCD* |
| 2024 | Msc Molecular Medicine | *TCD* |
| 2023-2027 | PhD Immunology (co-supervisor) | *TCD* |
| 2023 | BA(Mod) Molecular Medicine (intercalating Medical student) | *TCD* |
| 2022 | BA(Mod) Molecular Medicine (intercalating Medical student) | *TCD* |
| 2021-2022 | Bachelor in Medicine | *TCD* |
| 2021-2022 | Research Assistant | *TCD* |
| 2021-2022 | PhD student | *TCD* |
| 2021 | BA(Mod) Molecular Medicine (intercalating Medical student) | *TCD* |
| 2020-2021 | Mres Infection Biology | *UoM* |
| 2018-2019 | Mres Infection Biology | *UoM* |
| 2017-2021 | PhD Student (day-to-day supervisor, unofficial) | *TCD* |
| 2016 | Msc Immunology | *TCD* |
| 2015-2017 | PhD Student | *TCD* |
| 2015 | Msc Immunology | *TCD* |
| 2014 | Msc Immunology | *TCD* |

**Additional teaching duties** In 2016 I took on the academic duties for a Professor in TCD on teaching sabbatical. This included all aspects of the role, including practical supervision, assessment and marking, tutorials and administrative duties. I have organised visits for school students between 2013-2016. I sit on thesis committees for 3 PhD students.

## Outreach, organisation and PPI

**Discover Research Night**, Dublin Sep 2014. Marie Skłodowska-Curie-funded. I was the event organiser responsible for public engagement at Trinity Biomedical Sciences Institute, organising a team of 50 researchers in leading themed tours of scientific facilities. We were the most subscribed event of Dublin Research Night (>500 members of the public).

**10th International Symposium on Bordetella**, Sep 2013. Logistics manager responsible for operations of the conference including registration, AV equipment and support for speakers.

**Manchester Immunology Group Seminar Series**, May 2017 – Sep 2018. Co-lead organiser. Handling invitations, hospitality, communications, for renowned international speakers to visit University of Manchester.

**School talk**, Oaklands Community College, Edenderry Co. Offaly, 2021. “The life of a scientist”.

**Trinity Omics Core facility**. I have established formal pipelines for a single cell RNA-sequencing core service ‘Trinity Omics’. I have successfully lobbied my institute to fund a research assistant to support the service under my supervision.

**Patient involvement** I have worked alongside patient representatives for Vasculitis Ireland as part of the PARADISE project. I have pitched an idea to the St James hospital Patient Representative Group and received feedback on development of a research programme, lay abstract and patient information leaflet.

## Publications

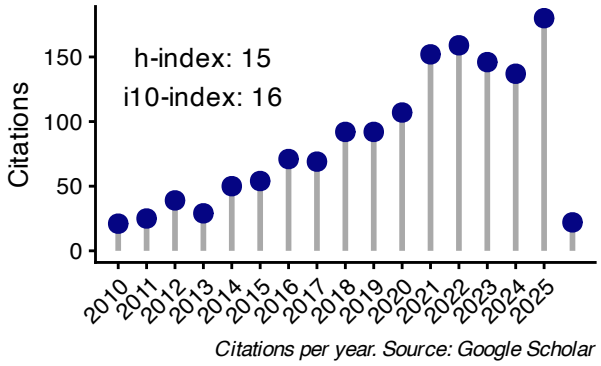
* (Co-)corresponding author or (co-)senior authorship


1. Ridge, K., Moran, B., Little, M. A., O’Farrelly, C., Dunne, J., Clowry, J., Loftus, C., Finlay, C. M., Irvine, A. D., & Conlon, N. (2025). Myeloid and Mast Cell Progenitors Are Elevated in Atopic Dermatitis. *JID Innovations*, *5*(5). `https://doi.org/10.1016/j.xjidi.2025.100390`
2. Hawerkamp, H. C., Yeow, A., Byrne, C. M., Chevalier, A., Matarazzo, L., Lawrence, A., Ivers, D., Murangi, T., O’Dowd, N., Hulshof, A.-M., Atiq, F., O’Sullivan, J. M., Kelly, V. P., Finlay, C. M., McSorley, H. J., Gangadharan, B., Reipert, B. M., O’Donnell, J. S., Turecek, P. L., & Fallon, P. G. (2025). The interleukin-33 receptor (ST2) is a novel therapeutic target to attenuate the progression of hemophilic arthropathy. *Blood*. `https://doi.org/10.1182/blood.2024027578`
3. Ridge, K., Ahmad, R., Moran, B., O’Farrelly, C., Dunne, J., Finlay, C. M., Irvine, A. D., & Conlon, N. (2025). Towards Personalised Therapy in Chronic Spontaneous Urticaria: Advancing From Endotype to Clinical Response. *Clinical & Experimental Allergy*, cea.70100. `https://doi.org/10.1111/cea.70100`

4. Suárez-Martins, M., González-Alayón, I., Casaravilla, C., Boon, L., Papotto, P. H., Finlay, C. M., Jenkins, S. J., Allen, J. E., & Díaz, Á. (2025). Macrophage IL-4 polarization is restricted by soluble CD40 agonists and antigen-induced CD154 but not by constitutive CD154 expressed by CD4+ T cells. *Journal of Leukocyte Biology*, *117*(7), qiaf069. https://doi.org/10.1093/jleuko/qiaf069
5. Dwivedi, A., Mhaonaigh, A. U., Carroll, M., Khosravi, B., Batten, I., Ballantine, R. S., Phelan, S. H., O'Doherty, L., George, A. M., Sui, J., Hawerkamp, H. C., Fallon, P. G., Noppe, E., Mason, S., Conlon, N., Cheallaigh, C. N., *Finlay, C. M., & *Little, M. A. (2024). Emergence of dysfunctional neutrophils with a defect in arginase-1 release in severe COVID-19. *JCI Insight*, *9*(17). https://doi.org/10.1172/jci.insight.171659
6. Ridge, K., Moran, B., Alvarado-Vazquez, P. A., Hallgren, J., Little, M. A., Irvine, A. D., O'Farrelly, C., Dunne, J., Finlay*, C. M., & Conlon*, N. (2024). Lin - CD117 + CD34 + Fc$\varepsilon$RI + progenitor cells are increased in chronic spontaneous urticaria and predict clinical responsiveness to anti- IgE therapy. *Allergy*. https://doi.org/10.1111/all.16127
7. Finlay*, C. M., & Allen*, J. E. (2024). IL-4-ever young: Type 2 cytokine signaling in macrophages slows aging. *Immunity*, *57*(3), 403–406. https://doi.org/10.1016/j.immuni.2024.02.013
8. Han, J., Gallerand, A., Erlich, E. C., Helmink, B. A., Mair, I., Li, X., Eckhouse, S. R., Dimou, F. M., Shakhsheer, B. A., Phelps, H. M., Chan, M. M., Mintz, R. L., Lee, D. D., Schilling, J. D., Finlay, C. M., Allen, J. E., Jakubzick, C. V., Else, K. J., Onufer, E. J., … Randolph, G. J. (2024). Human serous cavity macrophages and dendritic cells possess counterparts in the mouse with a distinct distribution between species. *Nature Immunology*, *25*(1), 155–165. https://doi.org/10.1038/s41590-023-01688-7
9. Tachó-Piñot, R., Stamper, C. T., King, J. I., Matei-Rascu, V., Richardson, E., Li, Z., Roberts, L. B., Bassett, J. W., Melo-Gonzalez, F., Fiancette, R., Lin, I.-H., Dent, A., Harada, Y., Finlay, C., Mjösberg, J., Withers, D. R., & Hepworth, M. R. (2023). Bcl6 is a subset-defining transcription factor of lymphoid tissue inducer-like ILC3. *Cell Reports*, *42*(11), 113425. https://doi.org/10.1016/j.celrep.2023.113425
10. Finlay, C. M., Parkinson, J. E., Zhang, L., Chan, B. H. K., Ajendra, J., Chenery, A., Morrison, A., Kaymak, I., Houlder, E. L., Murtuza Baker, S., Dickie, B. R., Boon, L., Konkel, J. E., Hepworth, M. R., MacDonald, A. S., Randolph, G. J., Rückerl, D., & Allen*, J. E. (2023). T helper 2 cells control monocyte to tissue-resident macrophage differentiation during nematode infection of the pleural cavity. *Immunity*, *56*(5), 1064–1081.e10. https://doi.org/10.1016/j.immuni.2023.02.016
11. McEntee, C. P., Houston, S., Finlay, C. M., Rossi, S., Liu, G., Shaw, T. N., Casulli, J., Fife, M., Smedley, C., Griffith, T. S., Pepper, M., Hussell, T., Hansbro, P. M., Schwartz, J.-M., Paidassi, H., & Travis, M. A. (2023). A subset of CD4+ effector memory T cells limit immunity to pulmonary viral infection and prevent tissue pathology via activation of latent TGF$\beta$. *bioRxiv*, 2023.03.02.527395. https://doi.org/10.1101/2023.03.02.527395
12. McManus, C. M., Bouchery, T., Suleiman, M., Kildemoes, A. O., Ferguson, A., Wang, T., Finlay, C. M., Chan, R., Renahan, T., Mukundan, A., Nkurunungi, G., & Bobardt, S. D. (2022). Hydra 2022: Return of the interactive conference on helminth parasitology after the pandemic. *Trends in Parasitology*, *38*(12), 999–1007. https://doi.org/10.1016/j.pt.2022.09.013
13. Finlay*, C. M., Parkinson, J. E., Chan, B. H. K., Ajendra, J., Chenery, A., Morrison, A., Houlder, E., Baker, S. M., Dickie, B., Boon, L., MacDonald, A., Konkel, J. E., Ruckerl, D., & Allen*, J. E. (2021). Genotype and Th2 Cells Control Monocyte to Tissue Resident Macrophage Differentiation During Nematode Infection of the Pleural Cavity. *bioXriv*, 2021.12.17.472661. https://doi.org/10.2139/ssrn.3992680
14. Fiancette, R., Finlay, C. M., Willis, C., Bevington, S. L., Soley, J., Ng, S. T. H., Baker, S. M., Andrews, S., Hepworth, M. R., & Withers, D. R. (2021). Reciprocal transcription factor networks govern tissue-resident ILC3 subset function and identity. *Nature Immunology*, *22*(10), 1245–1255. https://doi.org/10.1038/s41590-021-01024-x
15. Cunningham, K. T., Finlay, C. M., & Mills, K. H. G. (2021). Helminth Imprinting of Hematopoietic Stem Cells Sustains Anti-Inflammatory Trained Innate Immunity That Attenuates Autoimmune Disease. *Journal of Immunology (Baltimore, Md. : 1950)*, *206*(7), 1618–1630. https://doi.org/10.4049/jimmunol.2001225
16. Finlay, C. M., Cunningham, K. T., Doyle, B., & Mills, K. H. G. (2020). IL-33-Stimulated Murine Mast Cells Polarize Alternatively Activated Macrophages, Which Suppress T Cells That Mediate Experimental Autoimmune Encephalomyelitis. *Journal of Immunology (Baltimore, Md. : 1950)*, *205*(7), 1909–1919. https://doi.org/10.4049/jimmunol.1901321
17. Finlay*, C. M., & Allen*, J. E. (2020). The immune response of inbred laboratory mice to Litomosoides sigmodontis: A route to discovery in myeloid cell biology. *Parasite Immunology*, *42*(7), e12708. https://doi.org/10.1111/pim.12708
18. Czajkowska, B. I., Finlay, C. M., Jones, G., & Brown, T. A. (2019). Diversity of a cytokinin dehydrogenase gene in wild and cultivated barley. *PloS One*, *14*(12), e0225899. https://doi.org/10.1371/journal.pone.0225899
19. McEntee, C. P., Finlay, C. M., & Lavelle, E. C. (2019). Divergent Roles for the IL-1 Family in Gastrointestinal Homeostasis and Inflammation. *Frontiers in Immunology*, *10*, 1266. https://doi.org/10.3389/fimmu.2019.01266
20. Campbell, S. M., Knipper, J. A., Ruckerl, D., Finlay, C. M., Logan, N., Minutti, C. M., Mack, M., Jenkins, S. J., Taylor, M. D., & Allen, J. E. (2018). Myeloid cell recruitment versus local proliferation differentiates susceptibility from resistance to filarial infection. *eLife*, *7*, e30947. https://doi.org/10.7554/eLife.30947
21. Sutton, C. E., Finlay, C. M., Raverdeau, M., Early, J. O., DeCourcey, J., Zaslona, Z., O'Neill, L. A. J., Mills, K. H. G., & Curtis, A. M. (2017). Loss of the molecular clock in myeloid cells exacerbates T cell-mediated CNS autoimmune disease. *Nature Communications*, *8*(1), 1923. https://doi.org/10.1038/s41467-017-02111-0

22. Finlay*, C. M., Stefanska, A. M., Coleman, M. M., Jahns, H., Cassidy, J. P., McLoughlin, R. M., & Mills, K. H. G. (2017). Secreted products of Fasciola hepatica inhibit the induction of T cell responses that mediate allergy. *Parasite Immunology*, *39*(10), e12460. https://doi.org/10.1111/pim.12460
23. Finlay, C. M., Stefanska, A. M., Walsh, K. P., Kelly, P. J., Boon, L., Lavelle, E. C., Walsh, P. T., & Mills, K. H. G. (2016). Helminth Products Protect against Autoimmunity via Innate Type 2 Cytokines IL-5 and IL-33, Which Promote Eosinophilia. *Journal of Immunology (Baltimore, Md. : 1950)*, *196*(2), 703–714. https://doi.org/10.4049/jimmunol.1501820
24. Bernard, N. J., Finlay, C. M., Tannahill, G. M., Cassidy, J. P., O'Neill, L. A., & Mills, K. H. G. (2015). A critical role for the TLR signaling adapter Mal in alveolar macrophage-mediated protection against Bordetella pertussis. *Mucosal Immunology*, *8*(5), 982–992. https://doi.org/10.1038/mi.2014.125
25. Finlay, C. M., Walsh, K. P., & Mills, K. H. G. (2014). Induction of regulatory cells by helminth parasites: Exploitation for the treatment of inflammatory diseases. *Immunological Reviews*, *259*(1), 206–230. https://doi.org/10.1111/imr.12164
26. Coleman, M. M., Finlay, C. M., Moran, B., Keane, J., Dunne, P. J., & Mills, K. H. G. (2012). The immunoregulatory role of $CD4^+$ $FoxP3^+$ $CD25^-$ regulatory T cells in lungs of mice infected with Bordetella pertussis. *FEMS Immunology and Medical Microbiology*, *64*(3), 413–424. https://doi.org/10.1111/j.1574-695X.2011.00927.x
27. Walsh, K. P., Brady, M. T., Finlay, C. M., Boon, L., & Mills, K. H. G. (2009). Infection with a helminth parasite attenuates autoimmunity through TGF-beta-mediated suppression of Th17 and Th1 responses. *Journal of Immunology (Baltimore, Md. : 1950)*, *183*(3), 1577–1586. https://doi.org/10.4049/jimmunol.0803803

### Citation metrics




| Details | Number of papers |
|---|---|
| Total Papers | 29 |
| As First Author | 7 |
| Senior, Co-senior or primary supervisor | 4 |
| Seconday author - Major contribution | 9 |
| Seconday author - Minor contribution | 8 |
| As (co-)corresponding author | 5 |

## Referees

**Prof Judi Allen** — *University of Manchester*
Professor of Immunobiology; judi.allen@manchester.ac.uk — *Lydia Becker Institute*

**Prof Kingston HG Mills** — *Trinity College Dublin*
Professor of Experimental Immunology; kingston.mills@tcd.ie — *School of Biochemistry and Immunology*

**Prof Mark A Little** — *Trinity College Dublin*
Professor of Nephrology — *School of Medicine, Tallaght University Hospital*

**Dr Matthew Hepworth** — *University of Manchester*
Sir Henry Dale Fellow; matthew.hepworth@manchester.ac.uk — *Lydia Becker Institute*

**Prof Cliona O'Farrelly** — *Trinity College Dublin*
Professor of Comparative Immunology; ofarrecl@tcd.ie — *School of Biochemistry and Immunology*

# Appendix

## Skills

I have a broad skillset that includes animal research, translational research, molecular biology, bioinformatics and clinical data analysis that enables me to design and manage research projects from a multidisciplinary point of view.

| Category | Primary Skills | Tools & Technologies |
|---|---|---|
| Research & Analysis | Translational Research • Clinical Data Analysis • Animal Models • Patient Stratification | REDCap • Mass Cytometry • Flow Cytometry (30+ parameters) • ELISA |
| Programming | R (Fluent) • Python (Intermediate) • Unix/Bash (Basic) | RMarkdown • Git • HPC |
| Bioinformatics | Single Cell RNA-seq • Bulk RNA-seq • Mass Cytometry Analysis • Trajectory Analysis | Seurat • DESeq2 • limma • SCENIC • Spectre |
| Molecular Biology | NGS Library Prep • DNA/RNA Workflows • Proteomics • Cell Culture | Illumina NovaSeq • 10X Genomics • BD Rhapsody • TapeStation |
| Laboratory | Cell Sorting • Tissue Processing • Blood Processing • Core Facility Management | BD FACSAria • Mass Cytometry • Multiparameter Flow |

## Datasets, code & websites

**scRNA-seq data from Allergy paper 2024** — *NCBI*
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE261424 — *2024*

**scRNA-seq expression app** — *shiny app*
https://shiny.its.manchester.ac.uk/mdehsjpr/original/ — *2023*

**PARADISE R package for analysis of Kidney disease dataset (private)** — *R package*
On request — *2023*

**L sigmodontis collective analysis** — *Github*
https://github.com/Conorisco/Lito_database — *2022*

**scRNA-seq data from Immunity paper 2023** — *NCBI*
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE189031 — *2021*

## Short-listed Applications

- Short-listed for UKRI Innovation/Rutherford Fund Skills Development Fellowship (Computational Biology), 2017.
- Short-listed for academic-industry partnership post-doctoral position, Belgium, 2019
- Short-listed as candidate from school of medicine for Science Foundation Ireland Pathways programme, 2021
- Short-listed for Lister Prize 2023

## Training

| | | |
|---|---|---|
| 2023 | LAST (Ireland) course (rodents) | *LAST Ireland* |
| 2022 | Illumina Sequencing (NovaSeq 6000) | *Illumina* |
| 2018 | Single Cell RNA-sequencing | *Earlham Institute* |
| 2017 | Home Office personal animal licence | *Manchester* |
| 2014 | Fluorescent activated cell sorting | *UCD* |

## Thesis Committees and Continuation Vivas

| | | |
|---|---|---|
| 2025-2028 | School of Medicine PhD student | *Redacted* |
| 2025-2028 | School of Medicine PhD student | *Redacted* |
| 2022-2026 | School of Medicine PhD student | *Sarah Connolly* |
| 2022-2023 | School of Computer Science and Statistics PhD student | *Ultan Doherty* |
| 2021-2023 | School of Medicine PhD student | *Mariya Al Hamrashdi* |

## Interview Panels

| | | |
|---|---|---|
| 2022 | Post-Doctoral scientist at TCD (TTMI lab) | *Interview panel* |
| 2022 | Research Assistant at TCD (Core funded) | *Interview panel* |
| 2022 | Research Assistant at TCD (our lab) | *Interview panel* |
| 2021 | Research Assistant at TCD (our lab) | *Interview panel* |

## Courses

### University of Manchester

- Data Protection
- Introduction to High Performance Computing
- Introduction to version control using Git
- Introduction to the UNIX shell
- Programming with Python
- Data analysis using R
- Introduction to Python

### Trinity College Dublin

- Biological safety workshop
- Cryogenics safety workshop
- Radiological safety workshop
- LAST animal handling course (2009)
- Data Protection and Health Research (2021)
- Project Management
- Academic Practice: Perspectives on Teaching and Learning in Higher Education (1/3 SpCert - July 2022)
- Data Protection: Data Transfer and Secondary Use of Data (2021)
- Equality, Diversity and Inclusion in Higher Education

## Societies and Committees

### Society Membership

- British Society of immunology
- Irish Society of Immunology
- International Cytokine & Interferon Society
- Myeloid Network

### College Comittees

- STTAR Data Committee (2021-2023)
- Single Cell TCD core facility working group (Chair)
- FLOCI - Flow cytometry analysis committee (2021-Present)
- Post-Doc Representative for comparative medicine unit (CMU; animal research unit) executive committee (2016)
- Covid Strategic partnership (SFI Funded) (2021-2023)
- STTAR Covid-19 M-Bio LIMS (clinical data access, 2022)
- Trinity Omics (Acedemic Co-director and Committee Chair)
- TTMI Data Science Core
- Trinity Single Cell Omics Analysis Workshop